

Modeling and high-precision control of a ball-screw-driven stage

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Abstract

The demand for high-precision stages has received great attention due to the progress of nano-technology in recent years. Systems to provide long-range and high-precision performance for positioning, tracking and contouring actions have become stringent issues. Among these systems, the ball-screw-driven systems have been widely used in industrial applications and academic researches. In such systems, the friction behavior dominates the resulting performance and is usually known as the stick-slip phenomenon. The friction dynamics can be divided into the static and the dynamic regimes according to the conventional usage. In this paper, friction models are introduced to describe the dynamic behavior of a conventional ball-screw-driven *x*–*y* stage. The coherence between the theoretical and experimental data supports the validity of these models. Two sets of controllers corresponding to the static and the dynamic friction models are proposed based on the integral type sliding mode control (SMC) law. Experimental results demonstrate that the system achieves high-precision (10 nm) and long-range (10 cm) positioning performance with repeatability and robustness by the proposed control approaches.

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1. Introduction

Machining precision has always been an important issue in many engineering applications. The demands of high-precision performance have drastically increased in recent years due to the study of nano-science. Many researches have been conducted to characterize system behavior under micro/nano-scales. The study of friction behavior has attracted a great deal of interest due to its key role in high-precision motion applications. Conventionally, friction can be divided into two regimes: static friction and dynamic friction. Lots of work has been done to investigate friction dynamics [1–4] and different models have been proposed [5–7]. Investigations have revealed nonlinear microscopic behaviors of friction instead of the traditional macroscopic Coulomb description. Phenomena such as hysteresis, Stribeck effect, and plasticity have been reported. Nevertheless, their governing equations may be far from each other.

In control applications, compensation for friction is the paradigm to achieve high-precision performance. Significant steady state positioning error or tracking lag will exist if friction is not well compensated. In the control literature, Ro et al. [8] proposed a friction observer and a disturbance observer

based on a model [6]. Ro and Hubble [9] took a piece-wise linear reference model from a hysteretic torque–displacement curve in the controller design. Pan [10] used a PID controller design based on a model [7] for angular positioning. This thesis indicated the necessity of a holding force to maintain the desired position due to the spring-like nature of static friction. It also indicated spikes will be found in the positioning results due to the numerical quantization of measuring devices. Moreover, since the friction strongly depends on the contact conditions and are often characterized as the time and position dependency, Yang and Tomizuka [11] compensated for friction by adaptively changing the control pulse width. Tung et al. [12] compensated for friction with a repetitive control scheme. Others strategies such as the learning control in [13] and the fuzzy control in [14] have also been proposed in the literature.

In system consideration, a ball-screw-driven mechanism actuated by servo drives has widely been used in many applications due to its low cost and generality. The precision capabilities of different ball screws were investigated by Otsuka [15]. Other systems like the micro-positioning stages have also been studied in those years. Most studies have focused on the nano-scale positioning performance driven by piezo-actuators [16,17]. However, the limitation of operating range may become a major drawback for many applications.

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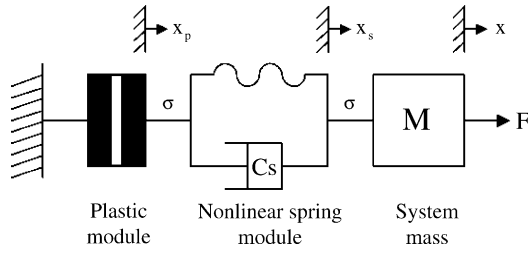


Fig. 1. The schematic concept of static friction model.

In this paper, positioning and tracking control based on the friction models are investigated. Different approaches are applied according to the dynamics of the static and the dynamic frictions. The controller designs are carried out by the sliding mode control (SMC) method in order to reject system disturbances and uncertainties. To avoid the chattering effect of conventional SMC, the integral type SMC is used instead. The performances of the system are demonstrated by using a ball-screw-driven servo system.

This article is organized as follows: in Section 2, two friction models are briefly introduced and the corresponding controller designs are described in Sections 3 and 4. The system setup and the experimental results with discussions are presented in Section 5. The conclusions are presented in Section 6.

2. Description of friction models

2.1. Static friction modeling

To describe the static friction dynamics, a novel static friction model proposed by Hsieh and Pan [7] is introduced. This model combines two major characteristics of static friction: the hysteresis and the plasticity. The conceptual description of this model as shown in Fig. 1 and consists of two modules: the nonlinear spring module and the plastic module. Due to

interactions between these two modules, a friction force σ will arise when an external force F is applied.

Consider the nonlinear spring module and let σ_s be the external force and x_s be the displacement of the spring. The governing equation of the nonlinear spring behavior is stated as:

$$\frac{d\sigma_s}{dx_s} = k_1 + k_2 e^{-\beta|x_s-x_r|} \quad (1)$$

where k_1 , k_2 , and β are positive scalars and x_s notes the active reverse point state. A combination of the above equation with the reverse point rules in [7] results in the general hysteresis behavior. The plastic module governs the behavior of plastic deformation, which is also reported in the literature of tribology. A state called the hardening state x_h is incorporated into the model in order to fully describe the plasticity. Let σ be the applied force, x_p be the plastic deformation. The governing equation of the plastic deformation is represented as:

$$\begin{cases} \dot{x}_h = \begin{cases} \alpha[f(|\sigma|) - x_h], & \text{if } f(|\sigma|) > x_h \\ 0, & \text{otherwise} \end{cases} \\ \dot{x}_p = \text{sgn}(\sigma)\dot{x}_h \end{cases} \quad (2)$$

where $f(|\sigma|) = |\sigma|^n/\lambda$ and α , λ , n are positive scalars.

Based on the hardening state, the entire system governing equation of static friction is divided into two phases: one for the system under creeping and the other when the system is completely hardened. In [7], the dynamic behavior of static friction is represented according to five state conditions, which might be inconvenient for further discussion and controller designs. A concise form can be obtained by introducing $\text{sgn}(\cdot)$ functions. The concise state equation is stated as follows.

Define the state vector by $[x_s \ \dot{x}_s \ x_h \ \dot{x}_h \ x_p \ \dot{x}_p]^T$, then the system described by Eqs. (1) and (2) can be represented by phase 1, for $|\sigma|^n/\lambda > x_h$, (the creep phase):

$$\frac{d}{dt} \begin{bmatrix} x_s \\ \dot{x}_s \\ x_h \\ \dot{x}_h \\ x_p \\ \dot{x}_p \end{bmatrix} = \begin{bmatrix} \dot{x}_s \\ \left(1 + \frac{\alpha n}{\lambda} |\sigma|^{n-1} C_s\right)^{-1} \left(\frac{F - \sigma}{m} - q\right) \\ \frac{\alpha |\sigma|^n}{\lambda} - \alpha x_h \\ \text{sgn}(\sigma) \left(1 + \frac{\alpha n}{\lambda} |\sigma|^{n-1} C_s\right)^{-1} \left(\frac{\alpha n}{\lambda} |\sigma|^{n-1} C_s \frac{F - \sigma}{m} + q\right) \\ \text{sgn}(\sigma) \left(\frac{\alpha |\sigma|^n}{\lambda} - \alpha x_h\right) \\ \left(1 + \frac{\alpha n}{\lambda} |\sigma|^{n-1} C_s\right)^{-1} \left(\frac{\alpha n}{\lambda} |\sigma|^{n-1} C_s \frac{F - \sigma}{m} + q\right) \end{bmatrix} \quad (3)$$

phase 2, for $|\sigma|^n/\lambda \leq x_h$, (the hardened phase):

$$\frac{d}{dt} \begin{bmatrix} x_s \\ \dot{x}_s \\ x_h \\ \dot{x}_h \\ x_p \\ \dot{x}_p \end{bmatrix} = \begin{bmatrix} \dot{x}_s \\ \frac{F - \sigma}{m} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

where

$$\sigma = C_s \dot{x}_s + k_1(x_s - x_r) + \text{sgn}(x_s - x_r) \frac{k_2}{\beta} (1 - e^{-\beta|x_s - x_r|}) + \sigma_r \quad (5)$$

$$q = \frac{\alpha n}{\lambda} |\sigma|^{n-1} (k_1 + k_2 e^{-\beta|x_s - x_r|}) \dot{x}_s - \text{sgn}(\sigma) \alpha \dot{x}_h \quad (6)$$

and C_s is the damping coefficient of the system. Then, the measured output displacement x is:

$$x = x_s + x_p \quad (7)$$

Combining the above equations with the reverse point rules, the response of system subjected to static friction can be obtained. The parameter values presented in the equations can be identified from the experimental data. The experimental system is the stage shown in Fig. 8 and the experiment is performed following the test method proposed in [7]. A typical experimental response of static friction compared with the theoretical data is shown together in Fig. 2. The result shows consistency between the real system and the model. In this study, experiments were conducted several times and the parameter values were varied w.r.t. position and time. This supports the time and position dependency reported in

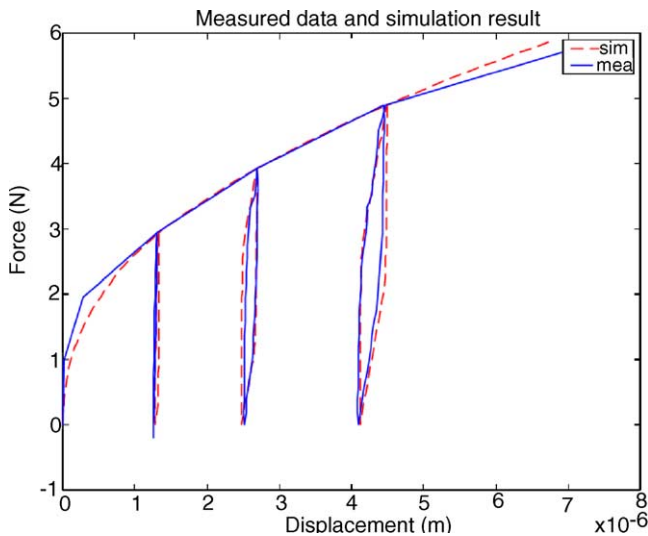


Fig. 2. A typical static friction response compared with simulation data.

Table 1
Parameter values of the static friction

k_1	6.12×10^6 (N/m)
k_2	2.88×10^8 (N/m)
β	1.11×10^8
n	2.283
α	1.64×10^{-2}
C_s	9.1×10^4
λ	9.15×10^6

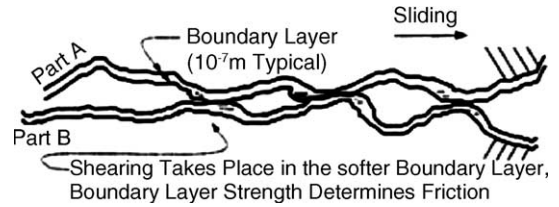


Fig. 3. The contact asperity model [4].

many investigations. A set of representative values is chosen and listed in Table 1 for further controller design usage.

In fact, the static friction behavior is similar to the mechanical structural behavior and can be explained by the asperity model [4] in Fig. 3. The contacting asperities between the upper and lower surfaces may join together due to cold welding and strong adhesion and form many tiny continuum-like structures called ‘micro-structures.’ These micro-structures are the source of static friction and they govern the static friction behavior. Since the micro-structures do not really bond together and might break if the external force is large to some extent. Hence, the asperity model can reasonably explain the static friction.

2.2. Dynamic friction modeling

To describe dynamic friction behavior, a friction model proposed by Canudas et al. [6] was adopted. The concept of this model is explained by the bristle model [8] shown in Fig. 4. The bristle deformation z is the source of friction force when contacting surfaces have relative velocity v . Then, the governing equation of bristle deformation is stated as:

$$\dot{z} = v - \frac{|v|}{g(v)} \sigma_0 z \quad (8)$$

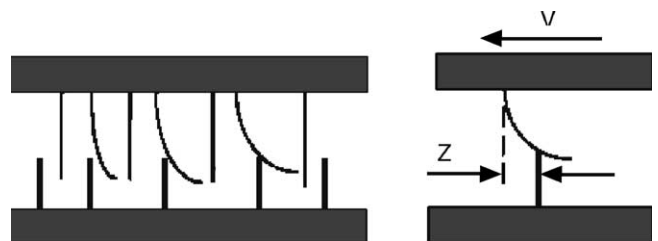


Fig. 4. The schematic concept of bristle model [8].

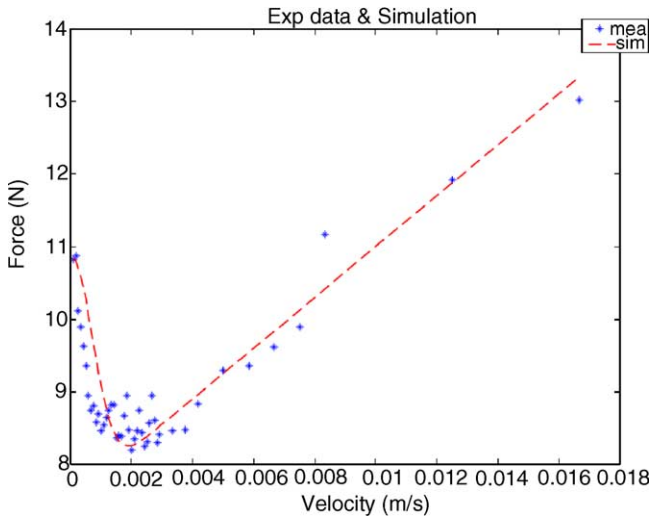


Fig. 5. A typical dynamic friction response compared with simulation data.

where

$$g(v) = F_C + (F_S - F_C)e^{-(v/v_s)^2} \quad (9)$$

is the nonlinear function presented in the governing equation and σ_0 denotes the bristle stiffness, F_C denotes the coulomb friction, F_S is the maximum static friction, and v_s denotes the Stribeck velocity defined in [6]. The general friction force F_f consisting of bending of the bristles, the damping effect, and the viscous friction is written as:

$$F_f = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v \quad (10)$$

where σ_1 and σ_2 denote the damping and the viscous coefficients, respectively.

In this model, the friction force is a function of velocity and the system dynamics under such friction load can be described by:

$$m\ddot{x} = F - F_f \quad (11)$$

where x denotes the displacement output of the system.

The parameter values of Eqs. (8)–(10) can be determined from the experimental data. The data were obtained by measuring the applying forces while the table was reached a constant velocity movement for various values. A typical experimental response compared with the theoretical response is shown in Fig. 5. The velocity information is constructed from the position data obtained in a high-resolution grid encoder. In the experiments, the identification is difficult to perform due to the strong position dependency. Even so, a characteristic curve of friction response can be extracted from the data in a statistical way and the identified parameter values are listed in Table 2.

Table 2

Parameter values of the dynamic friction

σ_0	1×10^5 (N/m)
σ_1	316 (N s/m)
σ_2	350 (N s/m)
F_S	10.8175 (N)
F_C	7.5 (N)
v_s	0.001 (m/s)

3. Sliding mode controller design for static friction regime

A controller design based on the static friction model is investigated in this section. Due to the complex nature of this model, a straight forward design is not possible. Instead, a nominal linear system is taken into consideration in this paper as stated by the following. From the relation of Eq. (7), it yields:

$$\dot{x} = \dot{x}_s + \dot{x}_p \quad \text{and} \quad \ddot{x} = \ddot{x}_s + \ddot{x}_p \quad (12)$$

Substituting Eq. (12) into Eqs. (3) and (4), a subsystem with only the measured states can be obtained in the following equations.

The creep phase:

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \dot{x}_s + \text{sgn}(\sigma) \left(\frac{\alpha |\sigma|^n}{\lambda} - \alpha x_h \right) \\ \frac{F - \sigma}{m} \end{bmatrix} \quad (13)$$

The hardened phase:

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \dot{x}_s \\ \frac{F - \sigma}{m} \end{bmatrix} \quad (14)$$

where σ is the same as in Eq. (5) and the second Newtonian Law is preserved in both cases. If a linear second order mass-damping-spring system (MCK) is used to nominally describe the output dynamics in Eqs. (13) and (14), it results in:

The creep phase:

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{m} & -\frac{C_s}{m} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F + \begin{bmatrix} \dot{x}_s + \text{sgn}(\sigma) \left(\frac{\alpha |\sigma|^n}{\lambda} - \alpha x_h \right) - \dot{x} \\ \frac{-\sigma + C_s \dot{x} + k_1 x}{m} \end{bmatrix} \quad (15)$$

The hardened phase:

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{m} & -\frac{C_s}{m} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F + \begin{bmatrix} 0 \\ \frac{-\sigma + C_s \dot{x} + k_1 x}{m} \end{bmatrix} \quad (16)$$

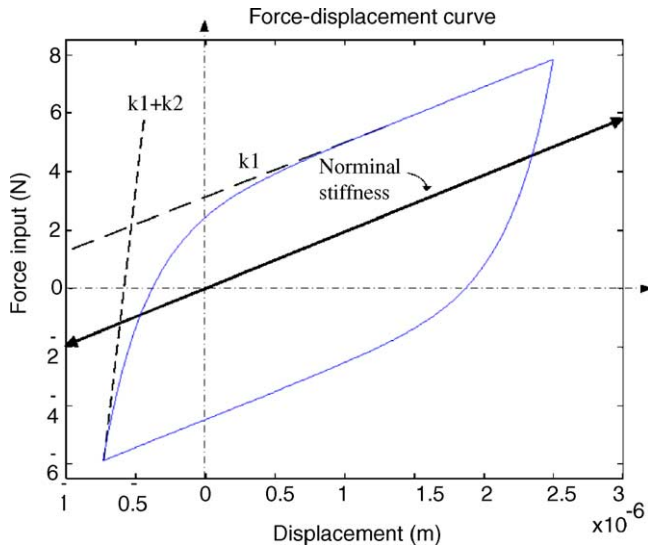


Fig. 6. Hysteresis represented by nonlinear stiffness.

where the stiffness constant and the damping coefficient of the nominal MCK are chosen to be the same values of k_1 and C_s in the friction model of Eqs. (5) and (6). The last right-hand term in each equation can be treated as the system disturbance or system deviation.

The merit of using Eqs. (15) and (16) is to consider the hysteresis behavior as a nonlinear stiffness constant with the nominal value k_1 as shown in Fig. 6. Due to the difficulties with state estimation of the system described by Eqs. (3)–(7), only the measured information is used for controller synthesis and it is more feasible in practical applications.

3.1. Sliding mode controller design

For tracking control purposes, the system equations in the form of error states are used. Define the error state as $e = x - x_d$, where x_d is the desired system displacement, and assume it to be twice differentiable w.r.t. time as:

$$\dot{e} = \dot{x} - \dot{x}_d \quad \text{and} \quad \ddot{e} = \ddot{x} - \ddot{x}_d \quad (17)$$

Substituting Eq. (17) into Eqs. (15) and (16) yields:

The creep phase:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} = & \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{m} & -\frac{C_s}{m} \end{bmatrix} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F \\ & + \begin{bmatrix} \dot{x}_s + \text{sgn}(\sigma) \left(\frac{\alpha |\sigma|^n}{\lambda} - \alpha x_h \right) - \dot{e} - \dot{x}_d \\ \frac{-\sigma + C_s \dot{e} + k_1 e}{m} - \ddot{x}_d \end{bmatrix} \end{aligned} \quad (18)$$

The hardened phase:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} = & \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{m} & -\frac{C_s}{m} \end{bmatrix} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F \\ & + \begin{bmatrix} 0 \\ \frac{-\sigma + C_s \dot{e} + k_1 e}{m} - \ddot{x}_d \end{bmatrix} \end{aligned} \quad (19)$$

Write the above equation in a general form as:

$$\frac{d}{dt} E = AE + Bu + D \quad (20)$$

where E , A , B , and D corresponds to the terms shown in Eqs. (18) and (19), respectively, then a sliding mode control law for the system described by Eq. (20) can be synthesized as follows:

Define the sliding surface s as:

$$s = SE = \begin{bmatrix} s_1 & s_2 \end{bmatrix} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} = s_1 e + s_2 \dot{e} \quad (21)$$

When system is in sliding or $s = 0$, then $\dot{e} = (s_1/s_2)e = -Ke$ where K is the ratio between s_1 and s_2 ; or simply let $s_2 = 1$ and $s_1 = K$. The equivalent control u_{eq} of the system can be obtained by satisfaction of the condition $\dot{s} = S\dot{E} = SAE + SBu_{eq} = 0$, then

$$u_{eq} = -(SB)^{-1}(SA)E = k_1 e + (C_s - mK)\dot{e} \quad (22)$$

The closed loop system equation when $s = 0$ becomes:

$$\dot{E} = AE + Bu_{eq} = (I_n - B(SB)^{-1}S)AE = A_{eq}E \quad (23)$$

where $A_{eq} = \begin{bmatrix} 0 & 1 \\ 0 & -K \end{bmatrix}$ denotes the equivalent system matrix and should be Hurwitz.

Define the switching control law to be in the following form:

$$u_{sw} = -(SB)^{-1}W \int \text{sgn}(s) dt \quad (24)$$

Thus, the sliding mode control law is synthesized by u_{eq} and u_{sw} as:

$$u = u_{eq} + u_{sw} = k_1 e + (C_s - mK)\dot{e} - mW \int \text{sgn}(s) dt \quad (25)$$

For a system subjected to disturbances, the relationship of $\dot{s} = S\dot{E} = -W \int \text{sgn}(s) dt + SD$ is implied and

$$s\dot{s}(-W \int \text{sgn}(s) dt + SD) \leq |s|(-W \int \text{sgn}(s) dt + SD) \quad (26)$$

Therefore, if $W \int \text{sgn}(s) dt > SD$ then the approaching condition $s\dot{s} < 0$ can be satisfied.

The reason for using the integral form instead of the conventional sign function in switching controls is mainly

following the suggestion in [10], where it was suggested that holding force in the static friction control due to the spring-like nature of the static friction is necessary. Generally, to hold a spring at some specific position, the applied force must hold and this force is provided by the integral term in Eq. (25). Besides, the actuator chattering effect can also be eliminated compared with the conventional SMC design with sign functions [18–20]. Due to the nature of the plastic module in Eq. (2), the creep motion will finally vanish if the applied force is not monotonic increasing or decreasing. Such results can either conclude that the controlled system starting from the creep phase will finally reach the hardened phase, or equivalently the mismatched disturbance will finally vanish and turn into matched disturbance. The matched disturbance then can be totally rejected by the sliding mode control and high-precision results can be achieved.

4. Sliding mode controller design for dynamic friction regime

In motion control applications, dynamic friction compensation schemes based on friction estimation are reported in many control studies. In this paper, an off-line friction estimation scheme based on the dynamic friction model in Eqs. (8)–(11) is utilized to pre-compensate the dynamic friction force as shown in Fig. 7. With such a strategy, the resulting system is described as follows.

Due to the estimation error, the system dynamics after forward compensation can be represented as:

$$m\ddot{x} = u - F_f + \hat{F}_f = u + \Delta F_f \quad (27)$$

where $\Delta F_f = \hat{F}_f - F_f$ denotes the friction estimation error. By defining $e = x - x_d$ and $\ddot{e} = \ddot{x} - \ddot{x}_d$ and assuming x_d is twice differentiable, Eq. (27) becomes:

$$m\ddot{e} = u + \Delta F_f - m\ddot{x}_d = u + d \quad (28)$$

where $d = \Delta F_f - m\ddot{x}_d$ is considered as the system disturbance. The state equation of Eq. (28) can be written as:

$$\frac{d}{dt} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e \\ \dot{e} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F + \begin{bmatrix} 0 \\ \Delta F_f - m\ddot{x}_d \end{bmatrix} \quad (29)$$

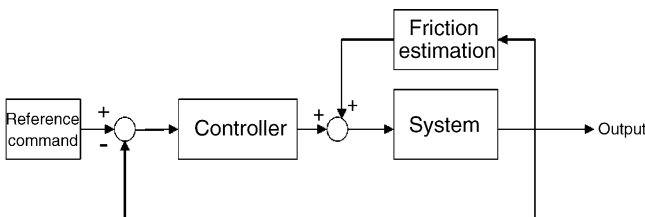


Fig. 7. The compensation scheme for dynamic friction.

When applying the conventional sliding mode control strategy, the equivalent control law will contain only the velocity feedback information due to the nature of the system matrix and may cause problems in implementation. In this paper, a controller design based on the method proposed by Chen and Lin [19] is introduced to avoid such problem. The design procedure is stated as follows. Define the sliding surface by:

$$s = \ddot{e} + K_v \dot{e} + K_p e \quad (30)$$

and let the sliding control law be:

$$u = u_{eq} + u_{sw} = m(\ddot{x}_d - K_v \dot{e} - K_p e) - mW \int \text{sgn}(s) dt \quad (31)$$

Substitute the above control law into Eq. (28) and from the definition of sliding surface, the closed-loop system equation can be obtained as:

$$\ddot{e} + K_v \dot{e} + K_p e = s = -W \int \text{sgn}(s) dt + \frac{1}{m} \Delta F_f \quad (32)$$

This reveals that the characteristic of the closed-loop system is now governed by $\ddot{e} + K_v \dot{e} + K_p e$. The corresponding approaching condition can be satisfied by:

$$\begin{aligned} s\dot{s} &= s \left(-W \text{sgn}(s) + \frac{1}{m} \Delta \dot{F}_f \right) \\ &= -W|s| + \frac{1}{m} s \Delta \dot{F}_f \leq |s| \left(-W + \frac{1}{m} \Delta \dot{F}_f \right) \leq 0 \end{aligned} \quad (33)$$

That is $W > \Delta \dot{F}_f / m$.

5. Experimental results

5.1. System description

The experimental system used in this study is a two-axis ball-screw-driven stage actuated by servo motors as shown in Fig. 8 associated with the control devices. A schematic diagram of a single axis stage is shown in Fig. 9. The stage displacement is measured by a grid encoder mounted on top of the table surface. A low-pass filter was introduced to reduce the noise effects and a 10 nm resolution could be obtained during operation. The control was implemented on an industrial computer with a sampling rate of 2000 Hz. The corresponding controller parameters are listed in Table 3.

5.2. Results and discussions

Two controllers specified as the static and the dynamic friction controllers were developed in the preceding sections. To achieve long-range capability, these two controllers were governed by a switching logic as shown in Fig. 10. The controllers were specifically assigned to the system according to

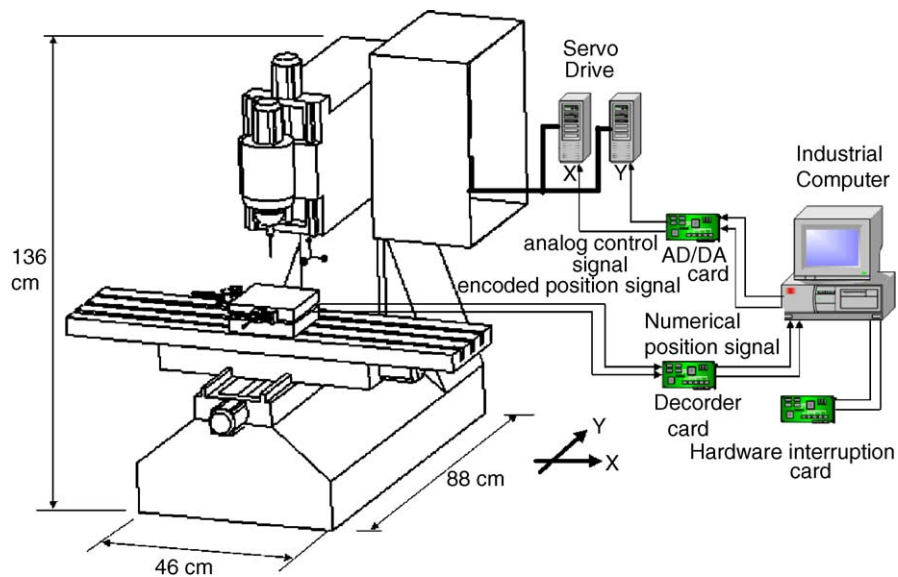


Fig. 8. The experimental system setup.

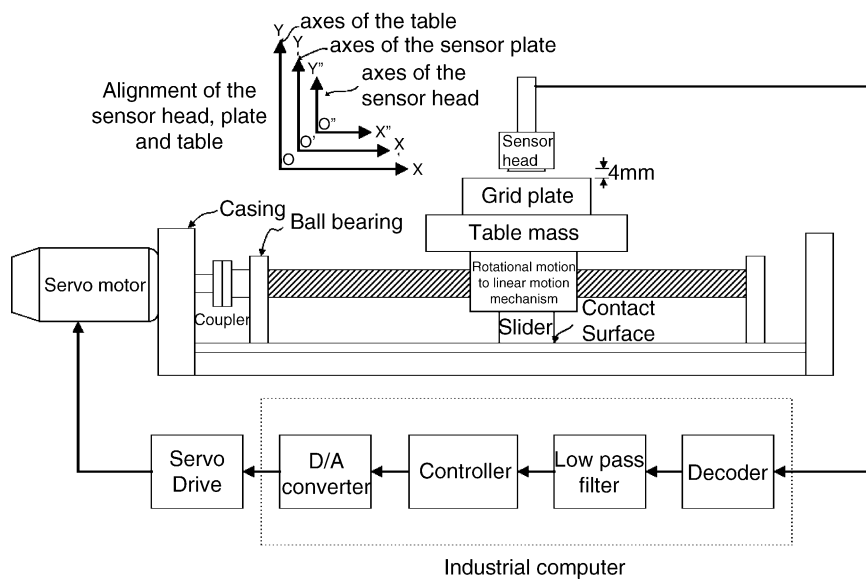


Fig. 9. The schematic diagram of single axis stage.

the friction regimes. The switching criterion of the two controllers is, in fact, a complex issue according to the reports in the tribology researches. Nevertheless, an empirical value of $1 \mu\text{m}$ was reported in many common ball-screw-driven

stage system experiments and this value is adopted for the switching criterion in the proceeding experiments. The results for positioning and tracking actions are stated in the following.

Table 3

Controller parameters

Static friction controller	
K	325
W	0.5
Dynamic friction controller	
K_p	1×10^5
K_v	1×10^3
W	10

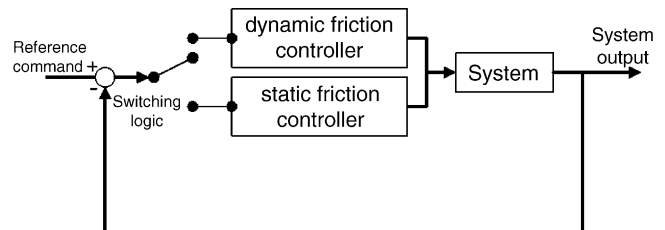


Fig. 10. The block diagram of positioning control.

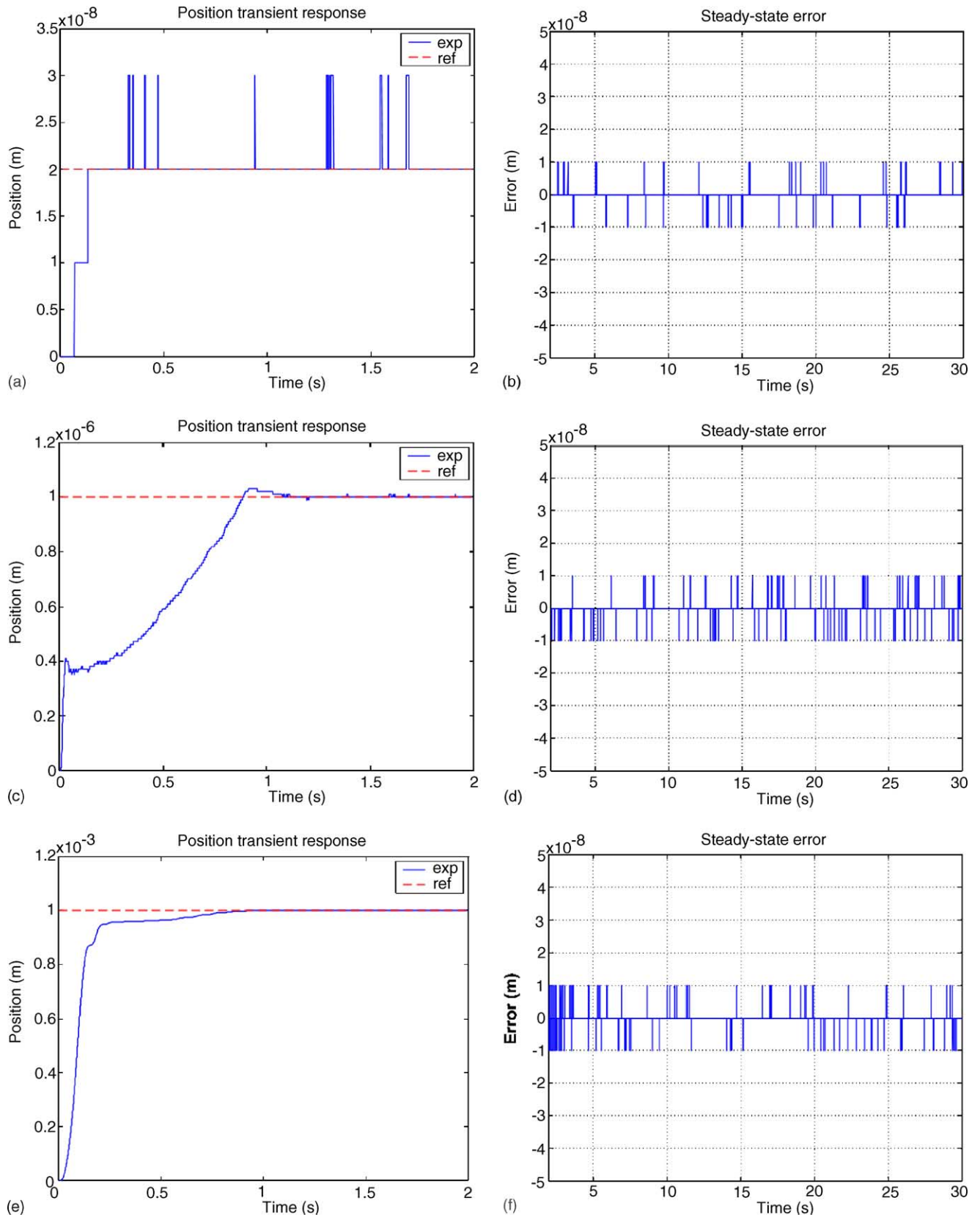


Fig. 11. The positioning responses. (a) The transient response to 20 nm command. (b) The steady-state error response of (a). (c) The transient response to $1 \mu\text{m}$ command. (d) The steady-state error response of (c). (e) The transient response to 1 mm command. (f) The steady-state error response of (e). (g) The transient response to 10 cm command. (h) The steady-state error response of (g).

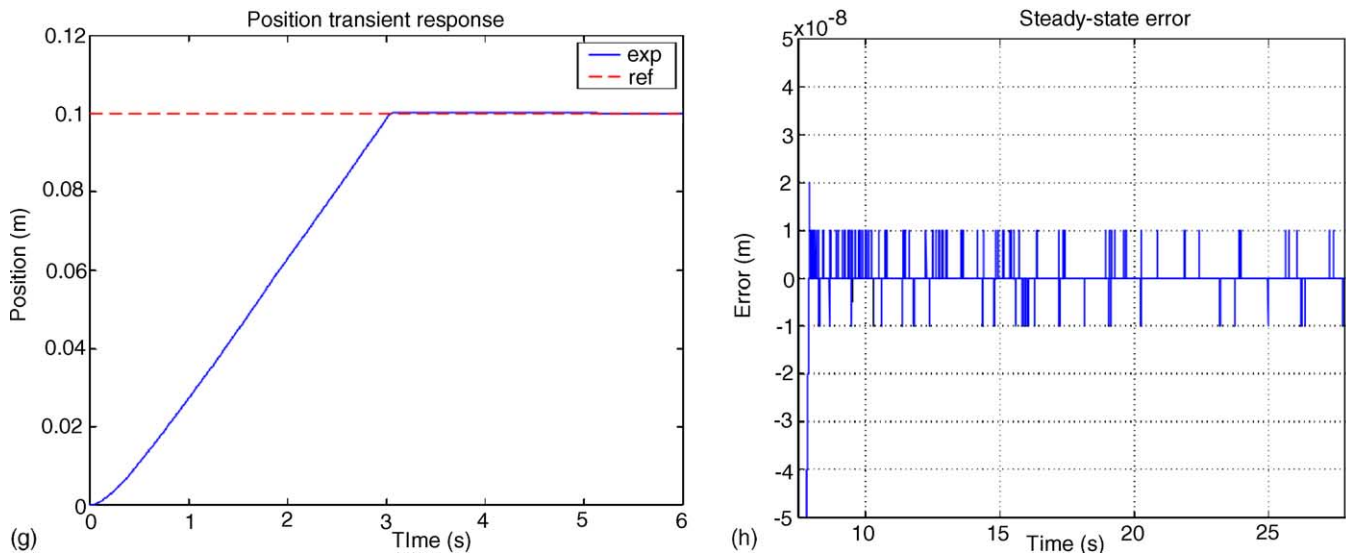


Fig. 11. (Continued).

For positioning control, reference commands ranging from 20 nm to 10 cm were conducted in the experiments. To clearly visualize the data, the results of transient responses and steady-state error responses are shown separately due to the large difference in data magnitudes. Results corresponding to 20 nm, 1 μ m, 1 mm, and 10 cm step command positioning are shown in Fig. 11(a–h), respectively. The steady-state error responses demonstrate good precision performance to ± 10 nm. The small peaks of 10 nm in magnitude may come from the spikes reported in [10] or from the noise of measurement devices and cannot be distinguished in our case. Even so, the precision performance does reach the resolution available in this system. To explore the repeatability of the positioning control, a stair-case reference

command was also performed in the experiments. The command was stepping 50 nm upward for every 1 s with a total of 300 steps and the overall displacement was up to 15 μ m. The result is presented with a partly enlarged figure and the steady-state error for every action is shown in Fig. 12. In the above results, the transient responses are found to be quite different. This can be explained in the following. Since the positioning distances are very different and due to the varying nature of the frictions, there will be uncertainties existing in the operation range. The sliding mode controller provides robustness only when system is in sliding. Hence the transient responses will be significantly affected by those uncertainties and various transient responses will be obtained.

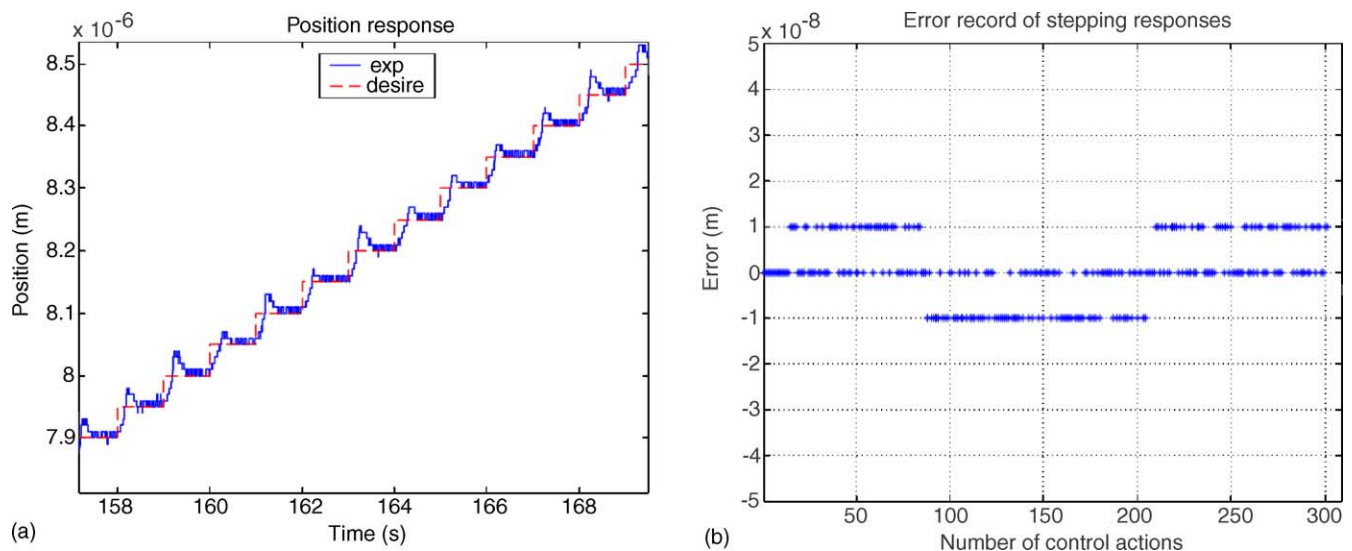


Fig. 12. The positioning response to a stepping reference command. (a) The position response (partly enlarged). (b) The steady state error.

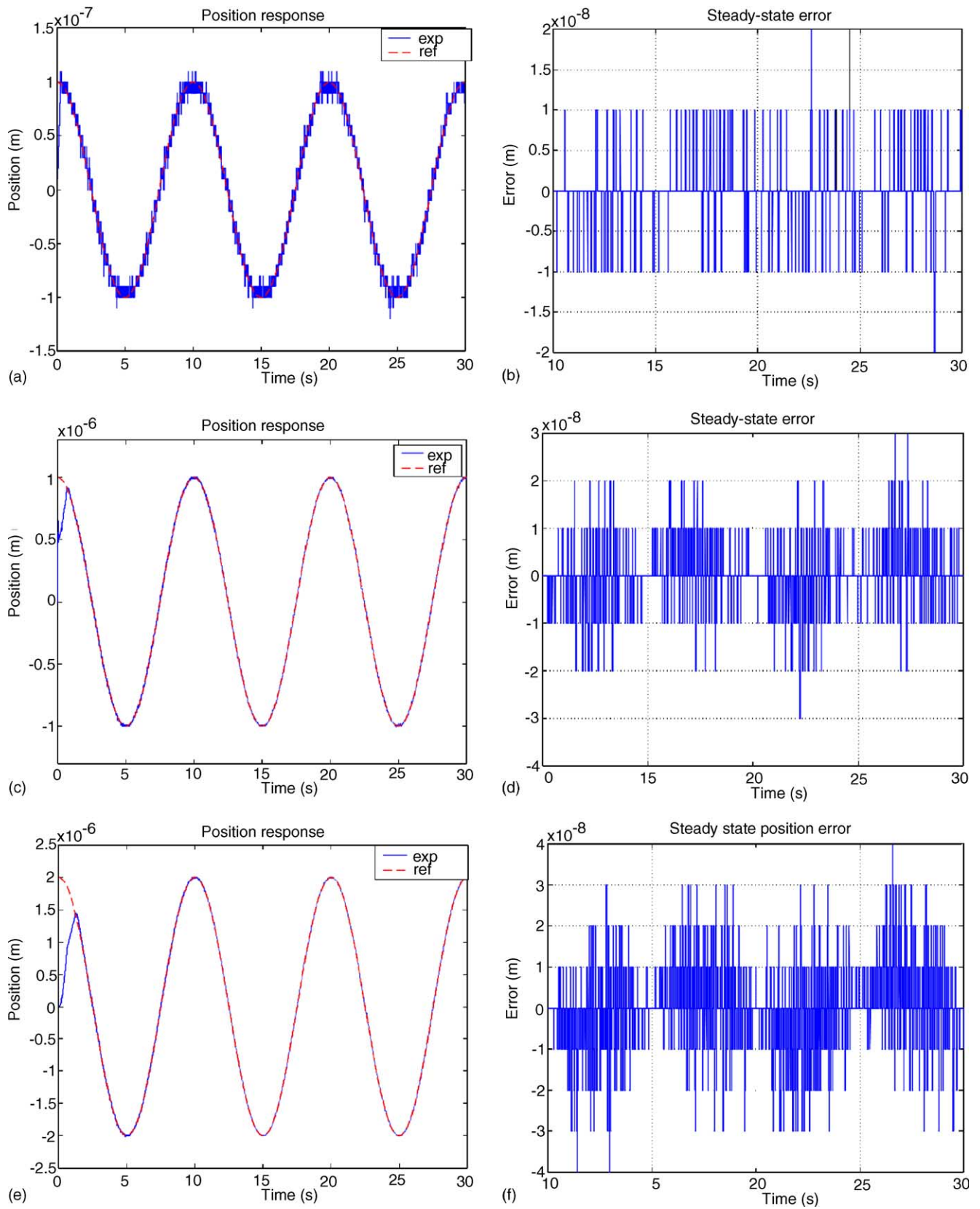


Fig. 13. Responses of tracking control to 0.1 Hz cosine signals. (a) The tracking response. (b) The error response of (a) after 10 s. (c) The tracking response. (d) The error response of (c) after 10 s. (e) The tracking response. (f) The error response of (e) after 10 s. (g) The tracking response. (h) The error response of (g) after 10 s.

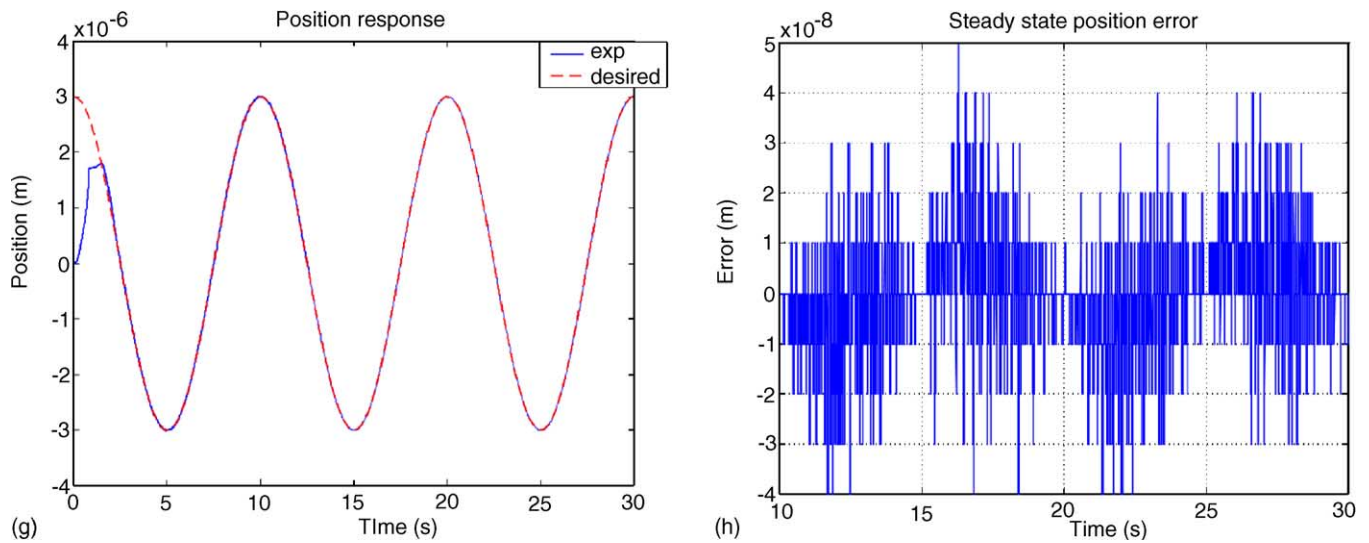
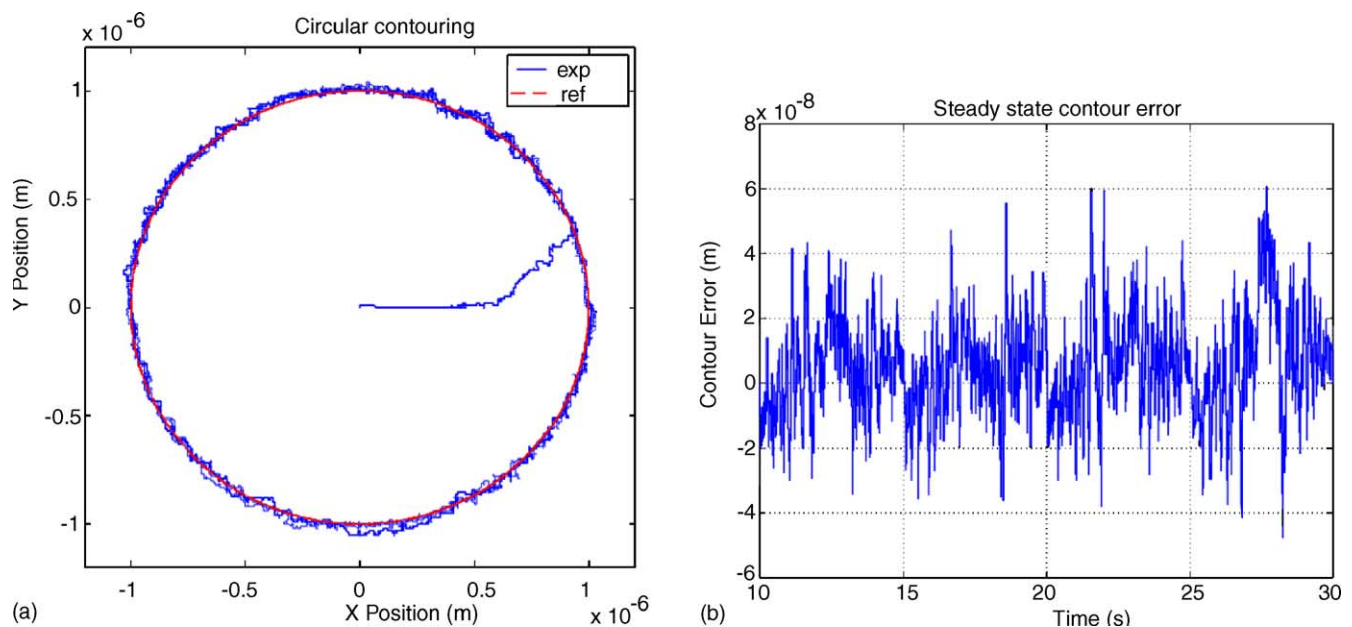


Fig. 13. (Continued).

For the tracking control study, 0.1 Hz cosine signals in different magnitudes ranging from 100 nm to 5 cm were performed in the system and shown in Fig. 13(a–h). Compared with the positioning actions, the results indicate that the hardened phase of a system in the static friction regime is not reached and a mismatched disturbance in Eq. (18) remains during the process. For a system in the dynamic friction regime, the estimation error in Eq. (27) is a time and position varying quantity due to the nature of friction. Although it is matched and can be rejected by the continuous time sliding mode control law, a precise friction estimation and the

infinite switching control action can not be achieved in practice. Therefore, the results of steady-state error responses were degraded as shown in Fig. 13(a–h).

Experimental results of the simultaneous tracking control of two axes were also shown in Figs. 14 and 15 corresponding to the circular and the square contours, respectively. This experiment is used to test the application of the controllers in contour plotting. The errors keep low while the range is short. The uncertainties between the two axes often degrade the contour performance and a more sophisticated contour control study is needed.

Fig. 14. The response of 0.1 Hz, $1 \mu\text{m}$ radius circular contour. (a) The position response of circular contour. (b) The contour error after 10 s.

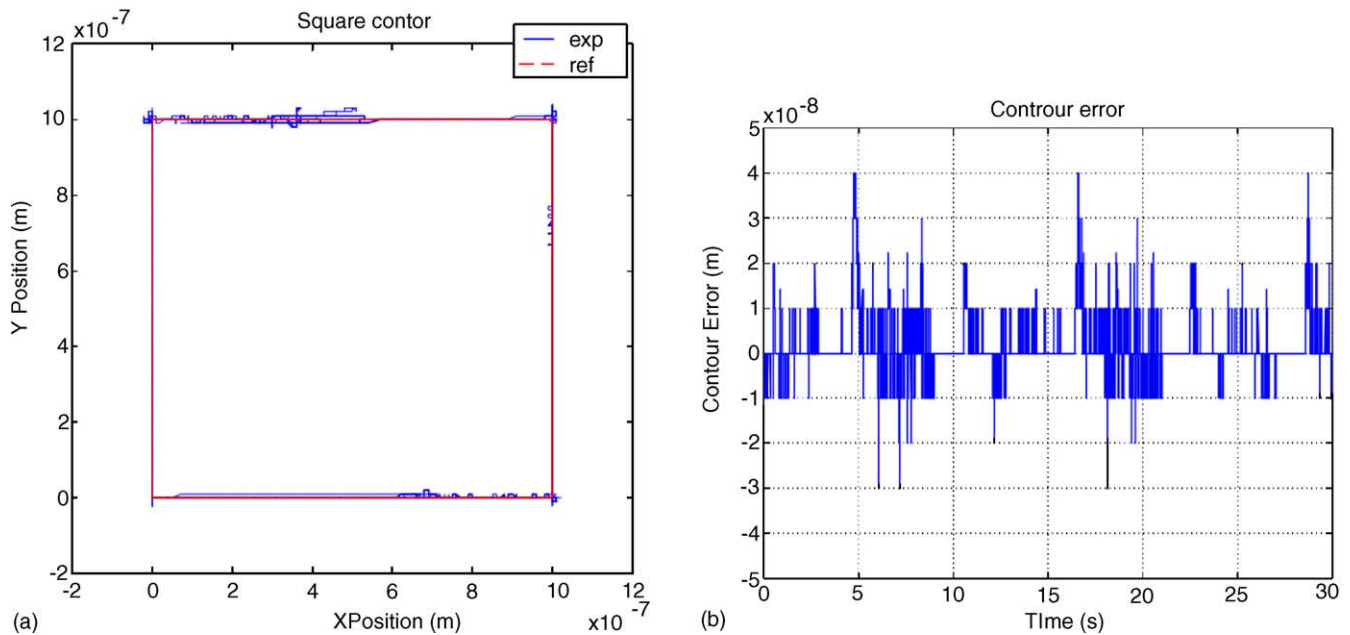


Fig. 15. The response of 0.1 Hz, 1 μ m square contour. (a) The position response of square contour. (b) The contour error.

6. Conclusions

In this paper, two friction models were adopted to describe the system behavior according to the static and dynamic friction regimes. Theoretical and experimental data of these models were in good coherence. A second-order linear MCK nominal model was proposed to simplify the highly nonlinear nature of the static friction. In addition, the creep and spring-like nature of the static friction was termed into mismatched and matched disturbances, respectively, that clearly characterize the creep and the hardened phases of static friction. To provide the system robustness, two integral sliding mode controllers were proposed based on different characteristics of the model dynamics. Comparing with the conventional SMC, the actuator chattering was eliminated by introducing an integral action. The controllers were activated according to the friction regimes to fulfill long-range operation. Experimental results of positioning control with high-precision and long-range requirements were achieved. Further investigation on improving the long-range tracking and contouring control will be held in the future.

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